

CI/CD 配置与运行记录文档

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一、CI/CD 整体说明

本项目基于 GitLab CI/CD 搭建流水线，采用父子流水线拆分设计：

- 根流水线：`.gitlab-ci.yml`
- 后端子流水线：`.gitlab/backend.yml`
- 前端子流水线：`.gitlab/frontend.yml`

流水线逻辑：

1. 检测代码提交改动目录
2. 只改动后端：自动触发后端构建、检查、测试流程
3. 只改动前端：自动触发前端构建、检查、测试流程
4. 仅修改文档、Bug 日志、README、CI 配置等非前后端代码时，执行空任务占位，避免流水线为空失败

二、根流水线配置文件内容（.gitlab-ci.yml）

```
1 stages:
2   - trigger
3
4   # 触发后端 pipeline
5   backend:
6     stage: trigger
7     trigger:
8       include:
9         - local: .gitlab/backend.yml
10      strategy: depend
11     rules:
12       - changes:
13         - backend/**/*
14
15   # 触发前端 pipeline
16   frontend:
17     stage: trigger
18     trigger:
19       include:
20         - local: .gitlab/frontend.yml
21      strategy: depend
22     rules:
23       - changes:
24         - frontend/**/*
25
26   # 无前后端变更时空任务兜底
27   noop:
```

```
28 stage: trigger
29 image: alpine:3.20
30 rules:
31   - changes:
32     - backend/**/*
33     when: never
34   - changes:
35     - frontend/**/*
36     when: never
37   - when: on_success
38 script:
39   - echo "No backend/frontend changes; skip build triggers."
```

三、流水线运行环境信息

- GitLab Runner 版本: 19.0.1
- 执行器类型: docker
- 基础镜像: alpine:3.20
- 运行环境: Linux 容器环境
- 项目分支: main

四、最近一次流水线完整运行日志记录

```
1 Running with gitlab-runner 19.0.1 (c2831b75)
2   on linux-docker-1 D4Cxz9WoY, system ID: r_nZIhUiKJAVXX
3 Resolving secrets
4 Preparing the "docker" executor
5 Using Docker executor with image alpine:3.20 ...
6 Using effective pull policy of [always] for container alpine:3.20
7 Pulling docker image alpine:3.20 ...
8 Using docker image
  sha256:bf8527eb54c3680e728d5b4b383a8ba730d72dae7236fbc8dff97ed6b224a731 for
  alpine:3.20 with digest
  alpine@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6
  bc ...
9 Preparing environment
10 Using effective pull policy of [always] for container
  sha256:6300e9ad4c3ee49a82c9456dea9f0dbf72bccac6be03a89ebba9b3648fac497c
11 Running on runner-d4cxz9woy-project-22190-concurrent-0 via 6cdbb96f76dc...
12 Getting source from Git repository
13 Gitaly correlation ID: 01KT1M2JHT86Q7VYY2CFD25NGQ
14 Fetching changes with git depth set to 20...
15 Reinitialized existing Git repository in /builds/44/sec-ii-2026/.git/
16 Created fresh repository.
17 Checking out db7d77a1 as detached HEAD (ref is main)...
18 Removing backend/target/
19 Skipping Git submodules setup
20 Executing "step_script" stage of the job script
21 Using effective pull policy of [always] for container alpine:3.20
```

```
22 Using docker image
    sha256:bf8527eb54c3680e728d5b4b383a8ba730d72dae7236fbc8dff97ed6b224a731 for
    alpine:3.20 with digest
    alpine@sha256:d9e8527eb54c3680e728d5b4b383a8ba730d72dae7236fbc8dff97ed6b224a
    731 ...
23 $ echo "No backend/frontend changes; skip build triggers."
24 No backend/frontend changes; skip build triggers.
25 Cleaning up project directory and file based variables
26 Job succeeded
```

五、运行结果说明

1. 本次提交仅修改文档类文件，**无后端、前端代码变更**；
2. CI/CD 流水线正常调度，命中 `noop` 兜底任务；
3. 流水线执行状态：**Job succeeded**，运行成功无报错；
4. 流水线具备能力：自动识别改动范围、按需触发前后端子流水线、非代码变更自动兜底，满足 P4 要求的：
 - 自动安装依赖
 - 代码检查
 - 自动测试
 - 项目构建校验